Sakila Database Mini Capstone Project

Project Overview

This Mini Capstone Project provides students with an opportunity to apply data analytics and SQL skills to a real-world business scenario using the Sakila database. Students will perform exploratory and analytical queries to generate actionable insights that support business decision-making.

Project Objective

The objective of this project is to conduct a comprehensive business analysis of the Sakila database by extracting and analyzing transactional data, identifying key performance trends, generating insights using SQL queries, and providing data-driven recommendations.

Part 1: Business Analysis Questions

1. Determine the total rental revenue generated by each store, including the earliest and latest payment dates.
2. Analyze the month-by-month revenue breakdown for each store.
3. Compare total rental revenue between stores and calculate percentage contribution.
4. Evaluate rentals per month and average payment per category.
5. Rank stores by revenue, rentals, and customer spending.
6. Identify highest revenue-generating film categories.
7. Determine top revenue-generating customers.
8. Analyze monthly rental trends by category.
9. Evaluate staff performance in rentals and revenue.
10. Provide strategic recommendations for revenue growth.

Part 2: SQL Query Development

1. Students will design SQL query prompt for each business question.
    a. Example: Business Question 1 - Determine the total rental revenue generated by each store, including the earliest and latest payment dates
    b. Query Prompt

1. Total Revenue and Reporting Period

Write a SQL query to calculate the **total revenue generated by each store**, including:

- Store ID

- Total revenue
- Earliest payment date
- Latest payment date

Use the payment table and connect payments to stores through the appropriate relationships.

2. Write the query - Each query must include a title, the question, SQL code, and explanation of logic.
   Example:

```sql
SELECT
    s.store_id,
    SUM(p.amount) AS total_revenue,
    MIN(p.payment_date) AS start_date,
    MAX(p.payment_date) AS end_date
FROM payment p
JOIN staff st ON p.staff_id = st.staff_id
JOIN store s ON st.store_id = s.store_id
GROUP BY s.store_id;
```

Each Business Question Query is available in the sakila_miniCapstone_businessquestions.sql file.

Part 3: Store Revenue and Performance Analysis Report

- Project Overview
- Database Description
- Business Objective
- Key Performance Indicators (KPIs)
- Analytical Approach
- Key Insights
- Strategic Recommendations

Deliverables

- Completed document with queries and analysis report.

Evaluation Criteria

- Accuracy of SQL Queries
- Depth of Analysis
- Clarity and Organization
- Use of Analytical Techniques
- Quality of Insights and Recommendations

Tools & Technologies

- MySQL / MySQL Workbench
- Sakila Database
- Word Processing Software